



Projectiles · Worksheet 1

Mechanics Chapter 7: Projectiles. Answer all questions. Show your working clearly. Total: 66 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 7. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ ms}^{-2}$. Treat each projectile as a particle moving freely under gravity, and give answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Horizontal projection	19	
B Projectiles launched at an angle	21	
C The trajectory equation and further problems	21	
Roar-level challenge	5	
Total	66	

Section A Horizontal projection

19 marks

1. A ball rolls off the edge of a horizontal table 0.8 m high. Find the time it takes to reach the floor. [2]

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2. A ball is thrown horizontally at 12 ms^{-1} from the top of a cliff 20 m high. [3]
Find the time it takes to reach the ground and the horizontal distance it travels.

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3. A stone is thrown horizontally from the top of a cliff 45 m high. It lands 30 m from the foot of the cliff. [3]
Find the speed at which the stone was thrown.

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4. A ball is thrown horizontally at 10ms^{-1} from a point 40 m above horizontal ground. [4]
Find the speed of the ball as it hits the ground, and the angle its path makes with the horizontal at that moment.

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5. A dart is thrown horizontally at 18 ms^{-1} towards a dartboard 2.4 m away. The dart is aimed directly at the bullseye, at the same height as the point of release.

(a) Find the time taken for the dart to reach the board. [1]

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(b) Find how far below the bullseye the dart hits the board. [2]

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(c) Find the speed at which the dart must be thrown horizontally to hit the board only 5 cm below the bullseye. [3]

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(d) State one modelling assumption you have used. [1]

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(Total for Question 5 is 7 marks)



Section B Projectiles launched at an angle

21 marks

6. A particle is projected at 20 ms^{-1} at 30° above the horizontal. Find the horizontal and vertical components of its initial velocity. [2]

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7. The particle in Question 6 is projected from horizontal ground. Find the time of flight and the range. [3]

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8. A ball is kicked from the ground at 25 ms^{-1} at 50° above the horizontal. Find the greatest height it reaches. [3]

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9. A golf ball is hit from horizontal ground at 30 ms^{-1} at 40° above the horizontal. Find the time of flight and the range of the ball. [4]

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10. A ball is kicked from a point O on horizontal ground at 24 ms^{-1} at an angle θ above the horizontal, where $\tan \theta = \frac{3}{4}$. A vertical wall stands 30 m from O , and the ball hits the wall.

(a) Find the time taken for the ball to reach the wall. [2]

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(b) Find the height above the ground at which the ball hits the wall. [3]

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(c) Determine whether the ball is rising or falling when it hits the wall. [2]

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(d) Find the speed of the ball when it hits the wall. [2]

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(Total for Question 10 is 9 marks)

**Section C The trajectory equation and further problems**

21 marks

11. A particle is projected from O at 10ms^{-1} at 45° above the horizontal. Show that the equation of its path is $y = x - 0.098x^2$. [2]

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12. Using the equation in Question 11, find the range of the particle on horizontal ground, and its height when $x = 5$. [3]

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13. A particle is projected from O with speed U at an angle θ above the horizontal. Show that its path has equation [3]

$$y = x \tan \theta - \frac{gx^2}{2U^2 \cos^2 \theta}.$$

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14. A ball is thrown from a point 1.5 m above horizontal ground at 15ms^{-1} at 35° above the horizontal. [4]
A fence 3 m high stands 12 m away, measured horizontally.
Determine whether the ball clears the fence, and by how much.

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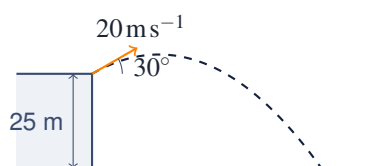
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15. A stone is thrown from the top of a vertical cliff, 25 m above the sea, with speed 20 m s^{-1} at 30° above the horizontal. It moves freely under gravity until it enters the sea.



- (a) Find the time taken for the stone to reach the sea. [3]

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- (b) Find the horizontal distance from the foot of the cliff to the point where the stone enters the sea. [2]

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- (c) Find the speed of the stone as it enters the sea, and the angle its direction of motion makes with the horizontal. [4]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A ball is kicked from horizontal ground at 21 m s^{-1} . It lands on the ground 40 m away. [5]
Find the two possible angles of projection, and state which one gives the shorter time of flight.

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END OF WORKSHEET

Total: 66 marks